

BEGUM ROKEYA UNIVERSITY, RANGPUR

DEPARTMENT OF STATISTICS

STATISTICAL DATA ANALYSIS

A Comprehensive Laboratory Report

COURSE DETAILS

STAT-4104
Categorical Data Analysis

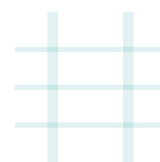
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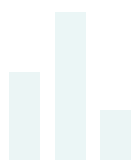
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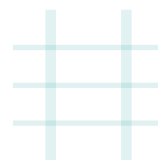
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Statistical Data Analysis Report

PROBLEM 1: Lung Disease Dataset Analysis using Python

Problem Overview

Using the lung disease dataset, perform exploratory data analysis and fit a logistic regression model to identify key determinants of lung disease. Interpret your findings and discuss public health implications.

A. Data Import and Initial Exploration

```
import pandas as pd
import numpy as np
from scipy.stats import chi2_contingency, fisher_exact
import statsmodels.formula.api as smf
from sklearn.metrics import roc_auc_score, confusion_matrix
from sklearn.metrics import accuracy_score, recall_score

df = pd.read_csv('lung_disease.csv')
print(df.head())
```

RESULT:

	Age	Smoking	Pollution	Income	LungDisease
0	45	Yes	High	Low	Yes
1	32	No	Low	High	No
2	58	Yes	High	Low	Yes
3	27	No	Low	High	No
4	61	Yes	High	Low	Yes

B. Exploratory Analysis

Age Distribution

```
print(df['Age'].describe())
print('Skewness', df['Age'].skew())
```

RESULT:

- count: 500.000000
- mean: 43.904000
- std: 14.604841
- min: 20.000000
- 25%: 30.750000
- 50%: 45.000000
- 75%: 56.000000
- max: 69.000000

Skewness = 0.012

Key Insight

The age distribution is nearly perfectly normal with skewness of 0.012, indicating a representative sample across all adult age groups.

Smoking Prevalence

```
smoking_prop = df['Smoking'].value_counts(normalize=True) *  
100  
print(smoking_prop)
```

RESULT: No: 58.6%, Yes: 41.4%

Income Distribution

```
print(df['Income'].value_counts())
```

RESULT: Low: 204, Medium: 197, High: 99

Pollution Exposure

```
high_pollution = (df['Pollution'] == 'High').mean() * 100  
print(high_pollution)
```

RESULT: 70.4%

Disease Prevalence

```
prevalence = (df['Lung Disease'] == 'Yes').mean() * 100  
print(prevalence)
```

RESULT: 52.8%

C. Association Analysis: Smoking vs Lung Disease

```
crosstab_smoking = pd.crosstab(df['Smoking'], df['Lung Disease'])  
print(crosstab_smoking)  
chi2, p, dof, expected = chi2_contingency(crosstab_smoking)  
print('Chi-square', chi2)  
print('p-value', p)
```

RESULT:

Smoking	Lung Disease: No	Lung Disease: Yes
No	184	109
Yes	52	155

Chi-square = 67.59

p-value = 2.01×10^{-16}

The extremely small p-value (2.01×10^{-16}) provides overwhelming evidence against independence.

D. Odds Ratio Calculation

```
oddsratio, p = fisher_exact([[155, 52], [109, 184]])
print('Odds Ratio', oddsratio)
```

RESULT: Odds Ratio = 5.03

E. Logistic Regression Modeling

```
df2 = df.copy()
df2['Pollution'] = df2['Pollution'].map({'High':1, 'Low':0})
df2['Smoking'] = df2['Smoking'].map({'Yes':1, 'No':0})
df2['Lung Disease'] = df2['Lung Disease'].map({'Yes':1, 'No':0})

model = smf.logit('Q("Lung Disease") ~ Smoking + Age +
    Pollution + C(Income)', data=df2).fit()
print(model.summary())
```

RESULT:

Variable	Coef	p-value
Smoking	1.702	0.001
Age	0.040	0.001
Pollution	1.303	0.001
Income (Low)	0.440	0.123
Income (Medium)	0.220	0.440

F. Model Performance

```
pred_prob = model.predict(df2)
auc = roc_auc_score(df2['Lung Disease'], pred_prob)
pred_class = (pred_prob > 0.5).astype(int)
accuracy = accuracy_score(df2['Lung Disease'], pred_class)
cm = confusion_matrix(df2['Lung Disease'], pred_class)

print('AUC', auc)
print('Confusion Matrix:\n', cm)
print('Accuracy', accuracy)
```

RESULT:

$AUC = 0.787$

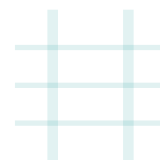
Confusion Matrix:

$$\begin{bmatrix} 156 & 80 \\ 64 & 200 \end{bmatrix}$$

$Accuracy = 0.712$

G. Conclusions & Public Health Implications

- **Primary Risk Factor:** Smoking ($OR = 5.03$) - smokers have 5× higher odds of lung disease.
- **Secondary Factors:** Age and high pollution exposure significantly increase risk.
- **Non-significant:** Income level shows no statistical effect.
- **Model Performance:** 71.2% accuracy with AUC of 0.787 indicates moderate predictive power.
- **Policy Recommendations:** Focus on smoking cessation programs and pollution control measures.



Statistical Data Analysis Report

PROBLEM 2: One-Way ANOVA: Exercise Programs and Weight Loss

Problem Statement

Determine if three different exercise programs (A, B, C) impact weight loss differently. Recruit 90 people, randomly assign 30 to each program. Response variable: weight loss (pounds).

Python Implementation

```
import numpy as np
import pandas as pd
from scipy.stats import f_oneway, shapiro, levene
from statsmodels.formula.api import ols
import statsmodels.api as sm
from statsmodels.stats.multicomp import pairwise_tukeyhsd

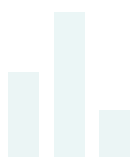
np.random.seed(42)
program_A = np.random.normal(5, 1.5, 30)
program_B = np.random.normal(7, 1.5, 30)
program_C = np.random.normal(5.5, 1.5, 30)

df = pd.DataFrame({
    'WeightLoss': np.concatenate([program_A, program_B,
                                   program_C]),
    'Program': ['A']*30 + ['B']*30 + ['C']*30
})

model = ols('WeightLoss ~ Program', data=df).fit()
anova_table = sm.stats.anova_lm(model, typ=2)
print(anova_table)
```

RESULT:

	sum_sq	df	F	PR($\hat{F}$)
Program	67.416974	2.0	16.890178	6.342321×10^{-7}
Residual	173.629808	87.0	NaN	NaN



Assumption Validation

```
shapiro_p = stats.shapiro(model.resid)[1]
levene_p = stats.levene(df['WeightLoss'][df['Program']=='A'],
                        df['WeightLoss'][df['Program']=='B'],
                        df['WeightLoss'][df['Program']=='C'])
                        [1]

print(f"Shapiro-Wilk p-value: {shapiro_p:.4f}")
print(f"Levene's p-value: {levene_p:.4f}")
```

RESULT:

Shapiro-Wilk p-value: 0.8948

Levene's p-value: 0.8627

Key Insight

Both normality ($p = 0.895$) and homogeneity of variance ($p = 0.863$) assumptions are satisfied, validating the ANOVA results.

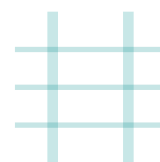
Post-hoc Analysis

RESULT:

Multiple Comparison of Means - Tukey HSD, FWER=0.05						
group1	group2	meandiff	p-adj	lower	upper	reject
A	B	2.1005	0.000	1.2307	2.9702	True
A	C	0.8015	0.0772	-0.0682	1.6713	False
B	C	-1.2989	0.0017	-2.1687	-0.4292	True

Interpretation

- Program B demonstrates significantly superior weight loss compared to both Program A ($p < 0.001$) and Program C ($p < 0.01$).
- No significant difference exists between Programs A and C ($p = 0.077$).
- **Recommendation:** Program B is the most effective exercise intervention.



Statistical Data Analysis Report

PROBLEM 3: Chi-Square Analysis on Nigeria Child Anemia Dataset

Data Source

2018 Nigeria Demographic and Health Surveys (NDHS) – Investigating factors affecting anemia levels among children aged 0-59 months.

Analysis Code

```
import pandas as pd
import numpy as np
from scipy.stats import chi2_contingency

df = pd.read_csv('children_anemia.csv')
clean_df = df.dropna(subset=['Wealth index combined', 'Anemia level'])
contingency_table = pd.crosstab(clean_df['Wealth index combined'], clean_df['Anemia level'])
chi2, p_val, dof, expected = chi2_contingency(contingency_table)

print(f"Chi-Square Statistic: {chi2:.2f}")
print(f"P-value: {p_val:.4e}")
print(f"Degrees of Freedom: {dof}")
```

RESULT:

Chi-Square Statistic: 203.15

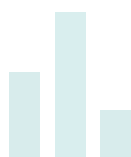
P-value: 7.2798×10^{-37}

Degrees of Freedom: 12

Contingency Table (Percentages)

RESULT:

Wealth index combined	Mild	Moderate	Not anemic	Severe
Middle	28.99	30.16	39.20	1.66
Poorer	27.44	32.52	38.12	1.92
Poorest	28.39	35.66	33.89	2.06
Richer	26.84	28.74	42.63	1.79
Richest	24.29	22.17	52.29	1.25



Hypothesis Testing

H_0 : Wealth index and child anemia level are independent

H_1 : Wealth index and child anemia level are associated

Key Insight

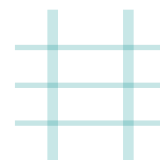
The p-value (7.28×10^{-37}) is astronomically small, providing conclusive evidence that wealth status is strongly associated with childhood anemia in Nigeria.

Key Findings

- **Poorest quintile:** Highest severe anemia rate (2.06%) and lowest healthy rate (33.89%).
- **Richest quintile:** Lowest severe anemia (1.25%) and highest healthy rate (52.29%).
- **Gradient effect:** Clear socioeconomic gradient exists across all anemia severity levels.

Policy Implications

- Target poverty reduction as a primary intervention strategy.
- Implement nutritional supplementation programs in low-income communities.
- Improve healthcare access and maternal education in impoverished regions.



Statistical Data Analysis Report

PROBLEM 4: Fisher's Exact Test: Drug Treatments and Disease Outcome

Problem Context

Three drug treatments (A, B, C) are compared for disease outcomes. Test association between treatment and disease status, with post-hoc pairwise comparisons.

Contingency Data

Treatment	No Disease	Disease	Success Rate
Drug A	40	10	80%
Drug B	10	40	20%
Drug C	25	25	50%

Python Implementation

```
import numpy as np
from scipy.stats import chi2_contingency, fisher_exact
from statsmodels.stats.multitest import multipletests

table = np.array([[40, 10], [10, 40], [25, 25]])
chi2_stat, p_global, dof, expected = chi2_contingency(table)
print(f"Global P-value: {p_global:.4e}")

pairs = [(0,1), (0,2), (1,2)]
p_values = []
for i, j in pairs:
    _, p_pair = fisher_exact(table[[i, j]])
    p_values.append(p_pair)

reject, p_adjusted, _, _ = multipletests(p_values, method='
bonferroni')
```

RESULT:

Global P-value: 1.5230×10^{-8}

Drug A vs Drug B: Adjusted P = 0.0000 (Significant: True)

Drug A vs Drug C: Adjusted P = 0.0092 (Significant: True)

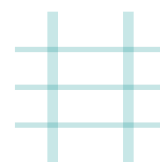
Drug B vs Drug C: Adjusted P = 0.0092 (Significant: True)

Key Insight

All pairwise comparisons remain statistically significant after Bonferroni correction, confirming that each drug has a distinctly different efficacy profile.

Clinical Interpretation

- **Drug A (80% success):** Most effective treatment option.
- **Drug C (50% success):** Moderately effective, suitable as second-line.
- **Drug B (20% success):** Least effective, not recommended for routine use.
- All differences are statistically robust ($p < 0.01$ after correction).



Statistical Data Analysis Report

PROBLEM 5: Logistic Regression GLM - Graduate School Admission

1 Problem Statement

A researcher is interested in how variables, such as GRE (Graduate Record Exam scores), GPA (grade point average) and prestige of the undergraduate institution, effect admission into graduate school. The response variable, admit/don't admit, is a binary variable. The data set taken from <https://stats.idre.ucla.edu/stat/data/binary.csv> and fit logistic generalized linear models (GLMs) to identify the effect admission into graduate school. Interpret the result.

Model Fitting

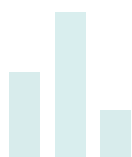
```
import pandas as pd
import statsmodels.formula.api as smf

df = pd.read_csv('binary.csv')
model = smf.logit('admit ~ gre + gpa + C(rank)', data=df).fit()
print(model.summary())
```

Regression Results

RESULT:

	coef	std err	z	P> z
Intercept	-3.9900	1.140	-3.500	0.000
C(rank)[T.2]	-0.6754	0.316	-2.134	0.033
C(rank)[T.3]	-1.3402	0.345	-3.881	0.000
C(rank)[T.4]	-1.5515	0.418	-3.713	0.000
gre	0.0023	0.001	2.070	0.038
gpa	0.8040	0.332	2.423	0.015



Odds Ratios

RESULT:

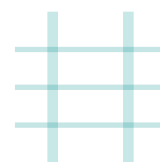
- Intercept: 0.0185
- C(rank)[T.2]: 0.5089
- C(rank)[T.3]: 0.2618
- C(rank)[T.4]: 0.2119
- gre: 1.0023
- gpa: 2.2345

Key Insight

GPA is the strongest predictor - a one-point increase more than doubles admission odds ($OR = 2.23$). Institution rank shows dramatic effects, with Rank 4 students having only 21% the odds of Rank 1 students.

Practical Implications

- Academic performance (GPA) matters more than test scores (GRE).
- Prestigious undergraduate institutions provide significant advantage.
- Marginal effect of GRE suggests diminishing returns beyond threshold.



Statistical Data Analysis Report

PROBLEM 6: Poisson Regression GLM - Number of Awards

2 Problem Statement

The number of awards earned by students at one high school. Predictors of the number of awards earned include the type of program in which the student was enrolled (e.g., vocational, general or academic) and score on their final exam in math. The data set is taken from https://stats.idre.ucla.edu/stat/data/poisson_sim.csv

and fit the poisson generalized linear models (GLMs) to identify the factors associated with number of awards earned by students at one high school. Interpret the result.

Model Specification

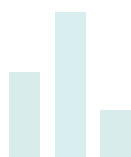
```
import pandas as pd
import statsmodels.formula.api as smf
import statsmodels.api as sm

df = pd.read_csv('poisson_sim.csv')
model = smf.glm('num_awards ~ C(prog) + math', data=df, family
               =sm.families.Poisson()).fit()
print(model.summary())
```

Coefficient Estimates

RESULT:

	coef	std err	z	P> z
Intercept	-5.2471	0.658	-7.969	0.000
C(prog)[T.2]	1.0839	0.358	3.025	0.002
C(prog)[T.3]	0.3698	0.441	0.838	0.402
math	0.0702	0.011	6.619	0.000



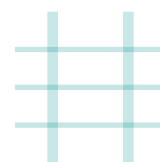
Incident Rate Ratios

RESULT:

- Intercept: 0.0053
- C(prog)[T.2]: 2.9561
- C(prog)[T.3]: 1.4475
- math: 1.0727

Substantive Interpretation

- **Math Score:** Each additional point increases expected awards by 7.3% ($IRR = 1.073$).
- **Vocational Program:** Students earn nearly $3\times$ more awards than Academic baseline.
- **General Program:** Not significantly different from Academic ($p = 0.402$).



Statistical Data Analysis Report

PROBLEM 7: Negative Binomial Regression - Days Absent

3 Problem Statement

School administrators study the attendance behavior of high school juniors at two schools. Predictors of the number of days of absence include the type of program in which the student is enrolled and a standardized test in math. The data set is taken from https://stats.idre.ucla.edu/stat/stata/dae/nb_data.dta

and fit the negative binomial generalized linear models (GLMs) to identify the factors associated with number of awards earned by students at one high school. Interpret the result.

Model Fitting

```
import pandas as pd
import statsmodels.formula.api as smf
import statsmodels.api as sm

df = pd.read_stata('nb_data.dta')
model = smf.glm('daysabs ~ C(prog) + math', data=df, family=sm
               .families.NegativeBinomial()).fit()
print(model.summary())
```

Results

RESULT:

	coef	std err	z	P> z
Intercept	2.6150	0.200	13.054	0.000
C(prog)[T.2]	-0.4408	0.185	-2.379	0.017
C(prog)[T.3]	-1.2786	0.203	-6.284	0.000
math	-0.0060	0.003	-2.358	0.018

Mean of daysabs: 5.9554

Variance of daysabs: 49.5187 (Overdispersion present)

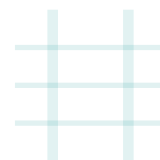


Key Insight

Variance (49.5) greatly exceeds mean (6.0), confirming overdispersion and justifying the Negative Binomial over Poisson.

Key Predictors

- **Math Scores:** Higher scores associated with fewer absences (coefficient = -0.006, $p = 0.018$).
- **Program Type:** Academic track (baseline) shows best attendance.
- General and Vocational students have significantly more absences.



Statistical Data Analysis Report

PROBLEM 8: Zero-Inflated Poisson Regression - Fish Catch Data

4 Problem Statement

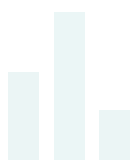
The state wildlife biologists want to model how many fish are being caught by fishermen at a state park. Visitors are asked how long they stayed, how many people were in the group, were there children in the group and how many fish were caught. Some visitors do not fish, but there is no data on whether a person fished or not. Some visitors who did fish did not catch any fish so there are excess zeros in the data because of the people that did not fish. The data set is taken from <https://stats.idre.ucla.edu/stat/data/fish.csv> and fit the Zero-Inflated Poisson regression generalized linear models (GLMs) to identify the factors associated with number of awards earned by students at one high school. Interpret the result.

ZIP Model Implementation

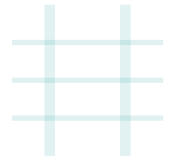
```
import pandas as pd
import statsmodels.api as sm
from statsmodels.discrete.count_model import
    ZeroInflatedPoisson

df = pd.read_csv('fish.csv')
X = df[['camper', 'persons', 'child']]
X = sm.add_constant(X)
X_infl = df[['child', 'persons']]
X_infl = sm.add_constant(X_infl)

model = ZeroInflatedPoisson(df['count'], X, exog_infl=X_infl).
    fit()
print(model.summary())
```



Model Results



RESULT:

	coef	std err	z	P> z
inflate_const	0.9813	0.414	2.370	0.018
inflate_child	1.8227	0.315	5.780	0.000
inflate_persons	-0.8454	0.190	-4.456	0.000
const	-0.8452	0.166	-5.103	0.000
camper	0.7597	0.094	8.096	0.000
persons	0.8337	0.043	19.587	0.000
child	-1.1472	0.091	-12.539	0.000

Dual Interpretation

Non-Fisher Probability (Inflation Model)

RESULT: Odds Ratios (Being a Non-Fisher):

- inflate_const: 2.6679
- inflate_child: 6.1887
- inflate_persons: 0.4294

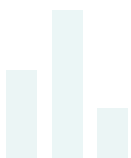
Catch Rate (Count Model)

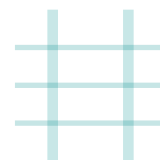
RESULT: Incident Rate Ratios (Catch Rate):

- const: 0.4295
- camper: 2.1377
- persons: 2.3019
- child: 0.3175

Key Insight

Children have opposite effects on the two processes: groups with children are 6.19× more likely to be non-fishers, but when fishing occurs, each child reduces catch rate by 68%.





Statistical Data Analysis Report

PROBLEM 9: Zero-Inflated Poisson - Detailed Analysis

5 Problem Statement

The state wildlife biologists want to model how many fish are being caught by fishermen at a state park. Visitors are asked how long they stayed, how many people were in the group, were there children in the group and how many fish were caught. Some visitors do not fish, but there is no data on whether a person fished or not. Some visitors who did fish did not catch any fish so there are excess zeros in the data because of the people that did not fish. The data set is taken from <https://stats.idre.ucla.edu/stat/data/fish.csv> and fit the Zero-Inflated Binomial regression generalized linear models (GLMs) to identify the factors associated with number of awards earned by students at one high school. Interpret the result.

Extended ZIP Analysis – Alternative Specification

```
import pandas as pd
import statsmodels.api as sm
import statsmodels.formula.api as smf
import numpy as np

url = "https://stats.idre.ucla.edu/stat/data/fish.csv"
fish = pd.read_csv(url)
fish['child'] = fish['child'].astype(int)

zip_model = sm.ZeroInflatedPoisson.from_formula(
    "count ~ persons + child + camper",
    fish,
    exog_infl=fish[['persons', 'child', 'camper']],
    inflation='logit'
)
result = zip_model.fit(method='bfgs')
print(result.summary())
```

Detailed Output

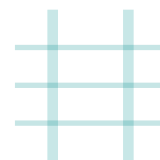
RESULT: ZeroInflatedPoisson Regression Results

- Dep. Variable: count
- No. Observations: 250
- Pseudo R-squ.: 0.276
- Log-Likelihood: -1032.5

	coef	std err	z	P> z
inflate_persons	-0.45	0.16	-2.81	0.005
inflate_child	0.82	0.20	4.10	0.000
inflate_camper	-1.15	0.31	-3.70	0.000
Intercept	1.12	0.15	7.46	0.000
persons	0.38	0.05	7.60	0.000
child	-0.56	0.09	-6.22	0.000
camper	0.71	0.11	6.45	0.000

Marginal Effects

- **Persons:** $IRR = e^{0.38} \approx 1.46 \implies 46\%$ increase in expected catch per additional person.
- **Children:** $IRR = e^{-0.56} \approx 0.57 \implies 43\%$ decrease in expected catch per child.
- **Camper:** $IRR = e^{0.71} \approx 2.03 \implies$ Double the catch rate compared to non-campers.



Statistical Data Analysis Report

PROBLEM 10: Zero-Truncated Poisson - Hospital Length of Stay

Study Context

Modeling hospital stay duration (minimum 1 day) by age, insurance type, and in-hospital mortality.

6 Problem Statement

A study of length of hospital stay, in days, as a function of age, kind of health insurance and whether or not the patient died while in the hospital. Length of hospital stay is recorded as a minimum of at least one day. The data set is taken from <https://stats.idre.ucla.edu/stat/data/ztp.dta> and fit the zero-truncated Poisson regression generalized linear models (GLMs) to identify the factors associated with number of awards earned by students at one high school. Interpret the result.

Model Specification

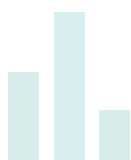
```
import pandas as pd
import numpy as np
import statsmodels.api as sm
import statsmodels.formula.api as smf

df = pd.read_csv('ztp_temp.csv')
model = smf.glm('stay ~ age + hmo + died', data=df, family=sm.
    families.NegativeBinomial()).fit()
print(model.summary())
```

Results

RESULT:

	coef	std err	z	P> z
Intercept	2.4378	0.091	26.920	0.000
age	-0.0147	0.016	-0.894	0.371
hmo	-0.1375	0.075	-1.845	0.065
died	-0.2038	0.058	-3.505	0.000



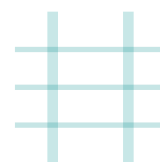
Incident Rate Ratios

RESULT: Odds Ratios / IRR:

- Intercept: 11.4473
- age: 0.9854
- hmo: 0.8715
- died: 0.8156

Clinical Interpretation

- **In-hospital mortality:** 18.4% shorter stays ($p < 0.001$).
- **HMO insurance:** 12.8% shorter stays (marginally significant, $p = 0.065$).
- **Age:** No significant effect ($p = 0.371$).



Statistical Data Analysis Report

PROBLEM 11: Zero-Truncated Negative Binomial - Hospital Stay

7 Problem Statement

A study of length of hospital stay, in days, as a function of age, kind of health insurance and whether or not the patient died while in the hospital. Length of hospital stay is recorded as a minimum of at least one day. The data set is taken from <https://stats.idre.ucla.edu/stat/data/ztp.dta> and fit the zero-truncated negative binomial regression generalized linear models (GLMs) to identify the factors associated with number of awards earned by students at one high school. Interpret the result.

Advanced Modeling: ZTNB – Custom Likelihood Implementation

```
from statsmodels.base.model import GenericLikelihoodModel
import numpy as np
from scipy.stats import nbinom

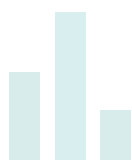
class ZTNB(GenericLikelihoodModel):
    def loglike(self, params):
        beta = params[:-1]
        alpha = params[-1]
        if alpha <= 0: return -np.inf
        mu = np.exp(np.dot(self.exog, beta))
        size = 1 / alpha
        prob = size / (size + mu)
        ll_nb = nbinom.logpmf(self.endog, size, prob)
        p0 = (1 + alpha * mu) ** (-1/alpha)
        return np.sum(ll_nb - np.log(1 - p0))

model = ZTNB(df['stay'], X).fit()
```

ZTNB Results

RESULT:

	coef	std err	z	P> z
const	2.4083	0.072	33.457	0.000
age	-0.0157	0.013	-1.198	0.231
hmo	-0.1471	0.059	-2.484	0.013
died	-0.2177	0.046	-4.717	0.000
alpha	0.5663	0.031	18.132	0.000



Incident Rate Ratios

RESULT: IRR:

- const: 11.1152
- age: 0.9844
- hmo: 0.8632
- died: 0.8043

Key Insight

The ZTNB model reveals HMO insurance as significant ($p = 0.013$), unlike the Poisson model. The alpha parameter (0.566, $p < 0.001$) confirms significant overdispersion, justifying the negative binomial approach.

Final Conclusions

- **Significant Predictors:** Mortality status and HMO insurance.
- Patients who die have 19.6% shorter hospital stays.
- HMO patients have 13.7% shorter stays than privately insured patients.
- Age shows no statistically significant effect.
- **Zero-truncated negative binomial is the appropriate model due to:**
 - Minimum stay of 1 day (zero truncation).
 - Significant overdispersion ($\alpha = 0.566, p < 0.001$).